

支持向量机

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$$\text{sign}(z) = \begin{cases} 1, & z \geq 0 \\ -1, & z < 0 \end{cases} \quad (0.1)$$

$$h_{\vec{w},b}(\vec{x}) = \text{sign}(\vec{w} \cdot \vec{x} + b) \quad (0.2)$$

$$\hat{\gamma}^{(i)} = y^{(i)}(\vec{w} \cdot \vec{x}^{(i)} + b) \quad (0.3)$$

$$\text{margin}(\vec{w}, b) = \min_{i=1, \dots, n} \frac{\hat{\gamma}^{(i)}}{\|\vec{w}\|} \quad (0.4)$$

几何边距:

$$\begin{aligned} \max_{\vec{w}, b} \quad & \text{margin}(\vec{w}, b) \\ \text{s.t.} \quad & \frac{1}{\|\vec{w}\|} \cdot y^{(i)}(\vec{w} \cdot \vec{x}^{(i)} + b) \geq \text{margin}(\vec{w}, b), \quad i = 1, \dots, n \end{aligned} \quad (0.5)$$

由于任意缩放 $\vec{w}$ 与 b 不会改变 $h_{\vec{w},b}(\vec{x})$ 的实际结果, 令 $\|\vec{w}\| = 1$, 函数边界:

$$\begin{aligned} \max_{\vec{w}, b} \quad & \text{margin}(\vec{w}, b) \\ \text{s.t.} \quad & y^{(i)}(\vec{w} \cdot \vec{x}^{(i)} + b) \geq \text{margin}(\vec{w}, b), \quad i = 1, \dots, n \\ & \|\vec{w}\| = 1 \end{aligned} \quad (0.6)$$

令 $\gamma = 1$

$$\begin{aligned} \min_{\vec{w}, b} \quad & \frac{1}{2} \|\vec{w}\|^2 \\ \text{s.t.} \quad & y^{(i)}(\vec{w} \cdot \vec{x}^{(i)} + b) \geq 1, \quad i = 1, \dots, n \end{aligned} \quad (0.7)$$

拉格朗日函数, 其中 $\alpha_i \geq 0$ 。

$$\mathcal{L}(\vec{w}, b, \vec{\alpha}) = \frac{1}{2} \|\vec{w}\|^2 - \sum_{i=1}^n \alpha_i [y^{(i)}(\vec{w} \cdot \vec{x}^{(i)} + b) - 1] \quad (0.8)$$

这里跳一步, 直接转化为对偶问题:

$$\max_{\vec{\alpha}} \min_{\vec{w}, b} \mathcal{L}(\vec{w}, b, \vec{\alpha}) \quad (0.9)$$

$$\nabla_{\vec{w}} \mathcal{L}(\vec{w}, b, \vec{\alpha}) = \vec{w} - \sum_{i=1}^n \alpha_i y^{(i)} \vec{x}^{(i)} = 0 \quad (0.10)$$

$$\frac{\partial}{\partial b} \mathcal{L}(\vec{w}, b, \vec{\alpha}) = \sum_{i=1}^n \alpha_i y^{(i)} = 0 \quad (0.11)$$

将等式0.10与等式0.11代入等式0.8得到：

$$\mathcal{L}(\vec{\alpha}) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n y^{(i)} y^{(j)} \alpha_i \alpha_j \langle \vec{x}^{(i)}, \vec{x}^{(j)} \rangle \quad (0.12)$$

$$\begin{aligned} \max_{\vec{\alpha}} \quad & \mathcal{L}(\vec{\alpha}) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n y^{(i)} y^{(j)} \alpha_i \alpha_j \langle \vec{x}^{(i)}, \vec{x}^{(j)} \rangle \\ \text{s.t.} \quad & \alpha_i \geq 0, i = 1, \dots, n \\ & \sum_{i=1}^n \alpha_i y^{(i)} = 0 \end{aligned} \quad (0.13)$$

$$\begin{aligned} \vec{w}^* &= \sum_{i=1}^n \alpha_i y^{(i)} \vec{x}^{(i)} \\ b^* &= -\frac{1}{2} (\max_{i:y^{(i)}=-1} \vec{w}^* \cdot x^{(i)} + \min_{i:y^{(i)}=1} \vec{w}^* \cdot x^{(i)}) \end{aligned} \quad (0.14)$$

$$\begin{aligned} \min_{\vec{w}, b} \quad & \frac{1}{2} \|\vec{w}\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.} \quad & y^{(i)} (\vec{w} \cdot \vec{x}^{(i)} + b) \geq 1, \quad i = 1, \dots, n \\ & \xi_i \geq 0, \quad i = 1, \dots, n \end{aligned} \quad (0.15)$$

$$\mathcal{L}(\vec{w}, b, \vec{\xi}, \vec{\alpha}, \vec{r}) = \frac{1}{2} \|\vec{w}\|^2 + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n \alpha_i \left[y^{(i)} (\vec{w} \cdot \vec{x}^{(i)} + b) - 1 \right] - \sum_{i=1}^n r_i \xi_i \quad (0.16)$$